Business Fundamentals: Quality Management

Himanshu Warudkar

Saturday, October 10th, 2026

From inspection to institution

Today's roadmap: Six Moves

ROADMAP



1 What is quality?

Garvin's lenses and dimensions; the 99.9% trap



2 Thinkers and variation

Shewhart to Taguchi



3 Economics of quality

Cost of quality; does quality pay? The evidence



4 Toolkits

TQM, Lean, Six Sigma, 7 tools, ISO and CMMI, DevOps/SRE



5 Cases: India and global

Dabbawalas, Aravind, UPI, Toyota, Boeing, CrowdStrike...



6 Synthesis

Takeaways and exit ticket

01

What is quality?

Everyone wants it. Few can define it, and that ambiguity is where strategy begins.

Warm-up: which is the higher-quality product?

ACTIVITY · 5 MIN



Maruti Suzuki Alto *vs* **Mercedes-Benz S-Class**



₹15 roadside cutting chai *vs* **₹350 café latte**



A ₹9,000 Android phone *vs* **A ₹1,50,000 flagship phone (it's a pro bro!)**

How to play

- Vote by show of hands for each pair.
- Two volunteers defend the minority view.
- Write down the word you used to justify your vote: price? reliability? feel? fit?

Five lenses on quality (Garvin, 1984)

CONCEPT



Transcendent

"I know it when I see it." Innate excellence; recognised through experience.

e.g. **Taj Mahal Palace, Mumbai**



Product-based

Quality is a measurable amount of an attribute (more is better).

e.g. **Thread count; camera megapixels**



User-based

Quality is fitness for use: whatever satisfies this customer's needs best.

e.g. **Alto for a first-time city buyer**



Manufacturing-based

Conformance to requirements: doing it right the first time.

e.g. **Six Sigma; zero defects**



Value-based

Performance at an acceptable price: quality relative to cost.

e.g. **Jio data plans; IndiGo**

Source: Garvin, D. A. (1984). What does "product quality" really mean? Sloan Management Review, 26(1), 25–43.

Eight dimensions: compete on a few, not all (Garvin, 1987)

CONCEPT

Performance

Primary operating characteristics

Car acceleration; app load time

Features

Secondary 'bells and whistles'

Sunroof; dark mode

Reliability

Probability of failure in a period

Mean time between failures; uptime

Conformance

Meets established specifications

Defect rate; SLA compliance

Durability

Product life before replacement

Godrej Storage

Serviceability

Speed, courtesy, ease of repair

Service network; ticket TTR

Aesthetics

Look, feel, sound, taste

Industrial design; UI polish

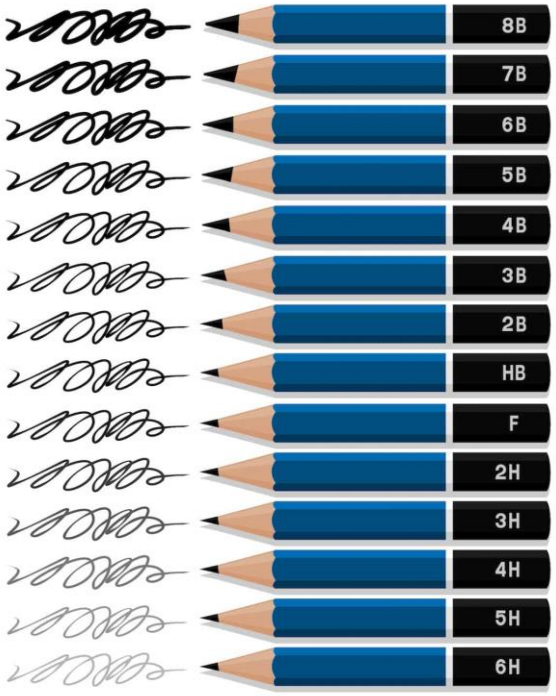
Perceived quality

Reputation, brand, inference

'Made in Japan' in the 1980s

Source: Garvin, D. A. (1987). Competing on the eight dimensions of quality. Harvard Business Review, 65(6), 101–109.

How do we define quality?







Dimension hunt: name a brand that wins on each dimension

ACTIVITY · 7 MIN · PAIRS

Performance	Brand: _____	**Features**	Brand: _____
Reliability	Brand: _____	**Conformance**	Brand: _____
Durability	Brand: _____	**Serviceability**	Brand: _____
Aesthetics	Brand: _____	**Perceived quality**	Brand: _____

Rules

- Include at least two Indian brands and two IT or digital products.
- For one brand, name the dimension it deliberately sacrifices.
- Fastest complete pair with a defensible trade-off wins.

Is 99.9% good enough? Do the maths at Indian scale

THINK FIRST · 3 MIN

Before the reveal, guess: how many failures per day would 99.9% success produce?

UPI payments

~698 million transactions/day (Dec 2025)

~6.98 lakh

failed payments/day at 99.9%

~2,400 at Six Sigma (3.4 per million)

Mumbai dabbawalas

~130,000 lunches delivered/day

~130

wrong or missed lunches/day at 99.9%

< 1 at Six Sigma (≈0.44/day)

Sources: NPCI data via Elets BFSI (Jan 2026): 21.63 bn UPI transactions in Dec 2025, ~698 mn/day. Thomke, S. (2012). Mumbai's models of service excellence. HBR, 90(11). Arithmetic is illustrative.

Great Product Poor Service Experience ...



02

The thinkers and the science of variation

Quality became a management discipline when we learned to separate signal from noise.

A century of quality: from inspection to system design

CONCEPT



Inspection and SQC

Shewhart's control chart at Bell Labs (1924)

Japan learns QC

Deming (1950) and Juran (1954) lecture to JUSE; Deming Prize (1951); Feigenbaum's Total Quality Control

Quality circles

Ishikawa's 7 tools and company-wide QC; Crosby's Quality Is Free (1979)

TQM in the West

Baldrige Award and ISO 9000 (both 1987); TQM goes mainstream

Six Sigma and Lean

Motorola (1986), GE (1995); TPS becomes 'Lean' (Womack et al., 1990); CMM/CMMI in software

Digital quality

DevOps, SRE error budgets, continuous testing; AI-assisted code raises new quality questions

Sources: Juran (1995) A History of Managing for Quality; Womack, Jones & Roos (1990) The Machine That Changed the World; ISO; NIST Baldrige Program.

The quality gurus: seven ideas every manager should own

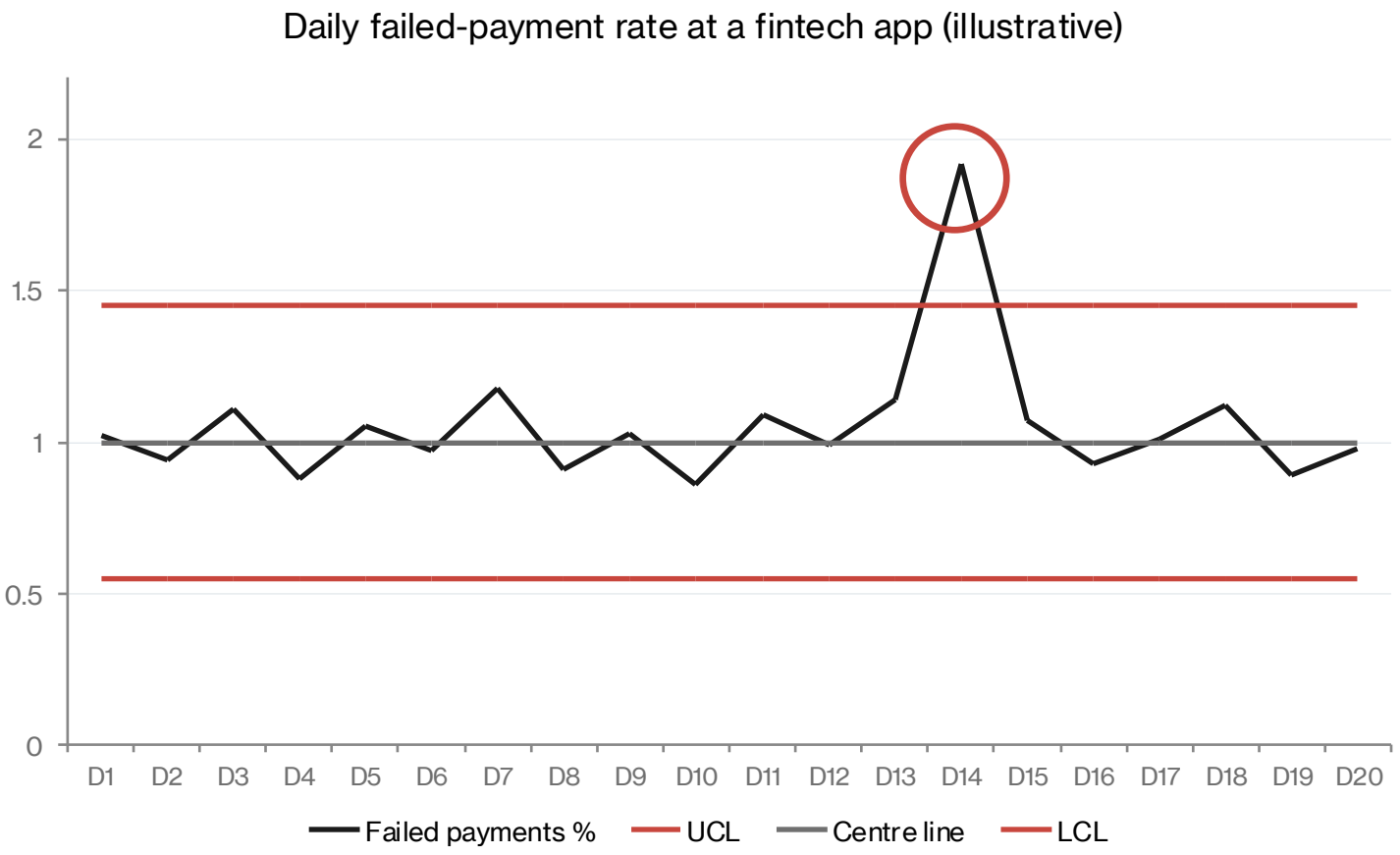
CONCEPT

Thinker	Big idea	Signature tool	Quality means…
Walter Shewhart	Variation has common and assignable causes	Control chart; Plan-Do-Check-Act origins	*a process in statistical control*
W. Edwards Deming	~94% of problems belong to the system, i.e. to management	PDCA cycle; 14 Points	*continual improvement of the system*
Joseph Juran	Quality trilogy: planning, control, improvement	Pareto: the 'vital few'	*fitness for use*
Armand Feigenbaum	Quality is everyone's job, across functions	Total Quality Control; cost of quality	*customer-defined, company-wide*
Philip Crosby	Quality is free; it is non-conformance that costs	Zero defects; price of non-conformance	*conformance to requirements*
Kaoru Ishikawa	Front-line workers can solve problems	Fishbone diagram; 7 tools; quality circles	*company-wide quality control*
Genichi Taguchi	Any deviation from target imposes a loss on society	Quality loss function; robust design	*minimum deviation from target*

Sources: Deming (1986) Out of the Crisis, p. 315; Juran (1986) Quality Progress; Crosby (1979) Quality Is Free; Feigenbaum (1951/1983) Total Quality Control; Ishikawa (1985); Taguchi (1986).

Signal or noise? Control charts tell the two apart

CONCEPT



Common cause

Inherent in the system: random, many small sources. Points stay within the limits.

→ **Fix the system (management's job)**

Special cause

Assignable: a bank's server went down on Day 14. Point breaks the UCL.

→ **Investigate and remove the specific cause**

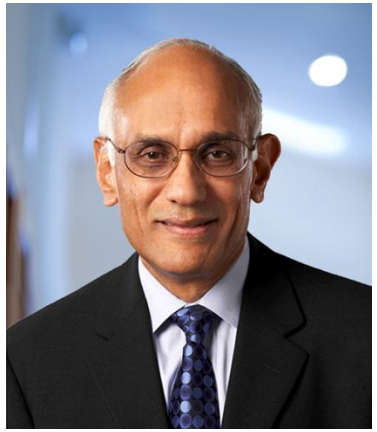
Tampering

Reacting to every wiggle as if it were special makes variation worse.

→ **Deming's funnel experiment**

Concept: Shewhart (1931) Economic Control of Quality of Manufactured Product; Deming (1986). Data are illustrative; limits set at ±3σ around the mean.

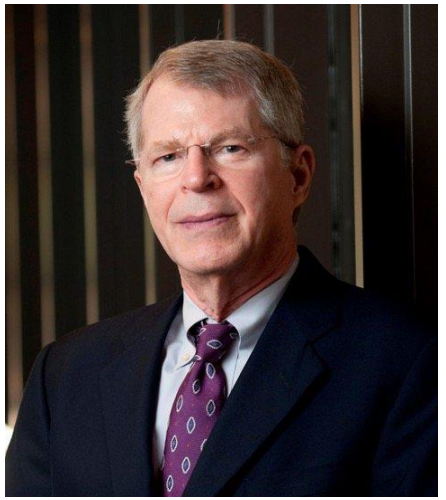
Introducing ServQual by PZB



A. Parasuraman

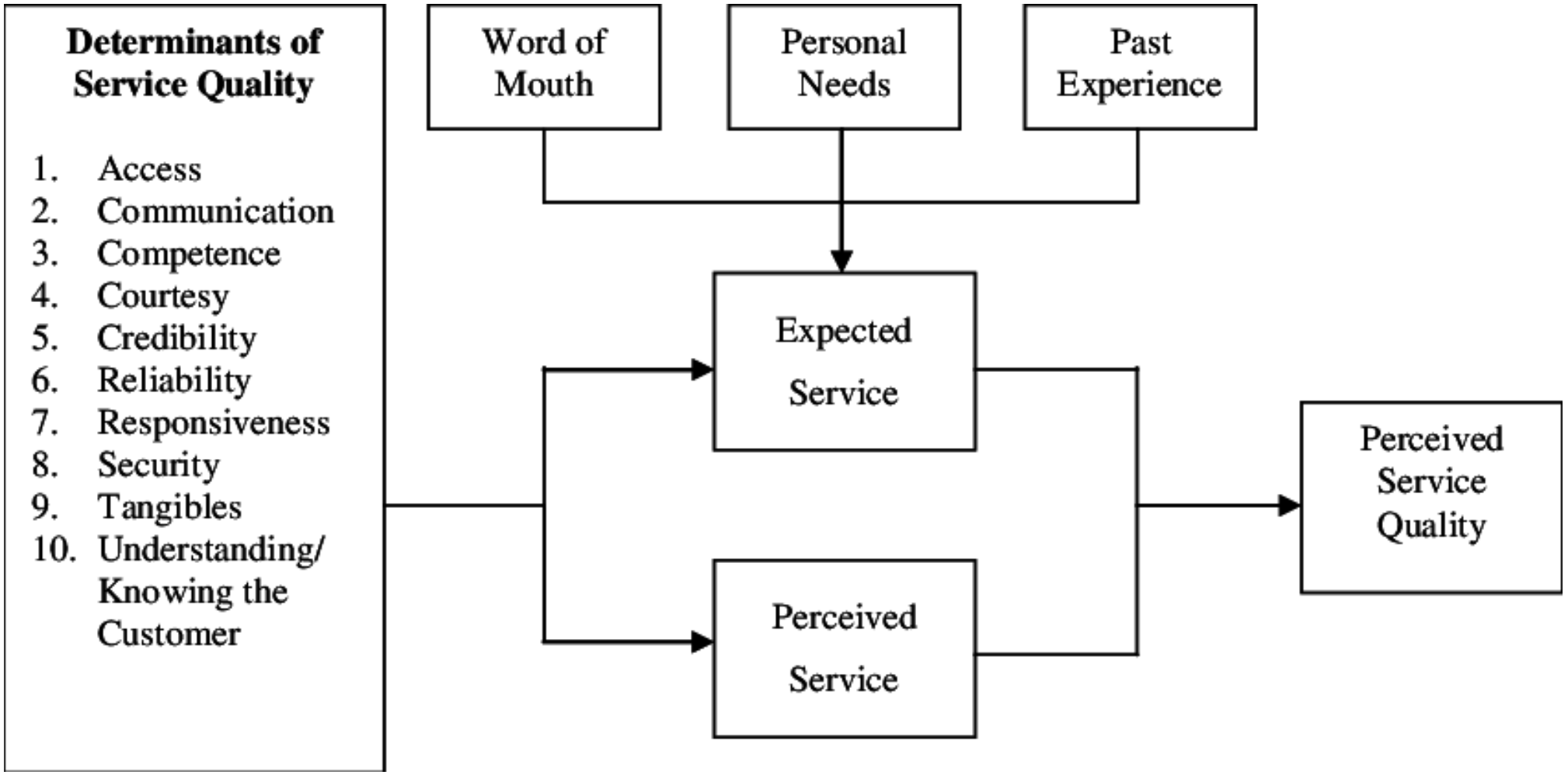


Valerie Zeithaml

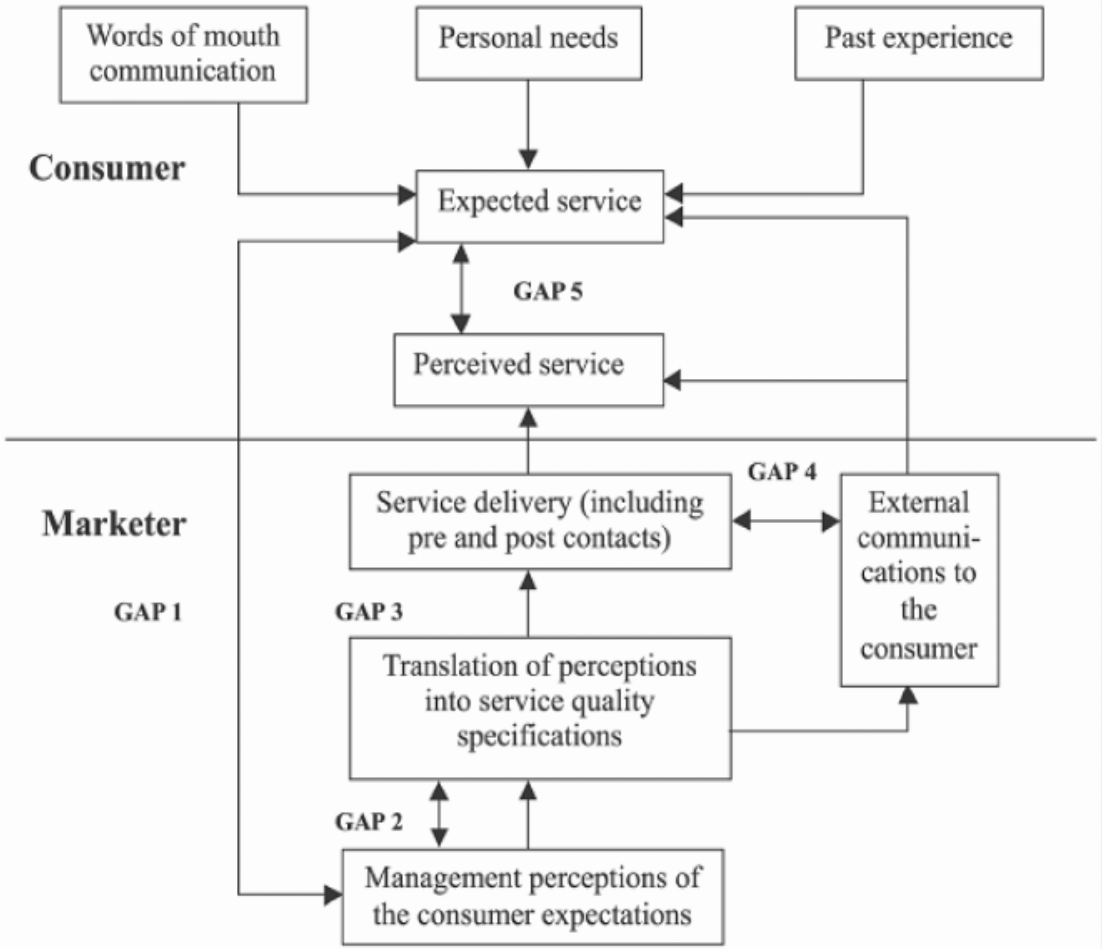


Leonard Berry

ServQual


Determinants of Service Quality
1. Access
2. Communication
3. Competence
4. Courtesy
5. Credibility
6. Reliability
7. Responsiveness
8. Security
9. Tangibles
10. Understanding/ Knowing the Customer
Word of Mouth
Personal Needs
Past Experience
Expected Service
Perceived Service
Perceived Service Quality

The Gap Model

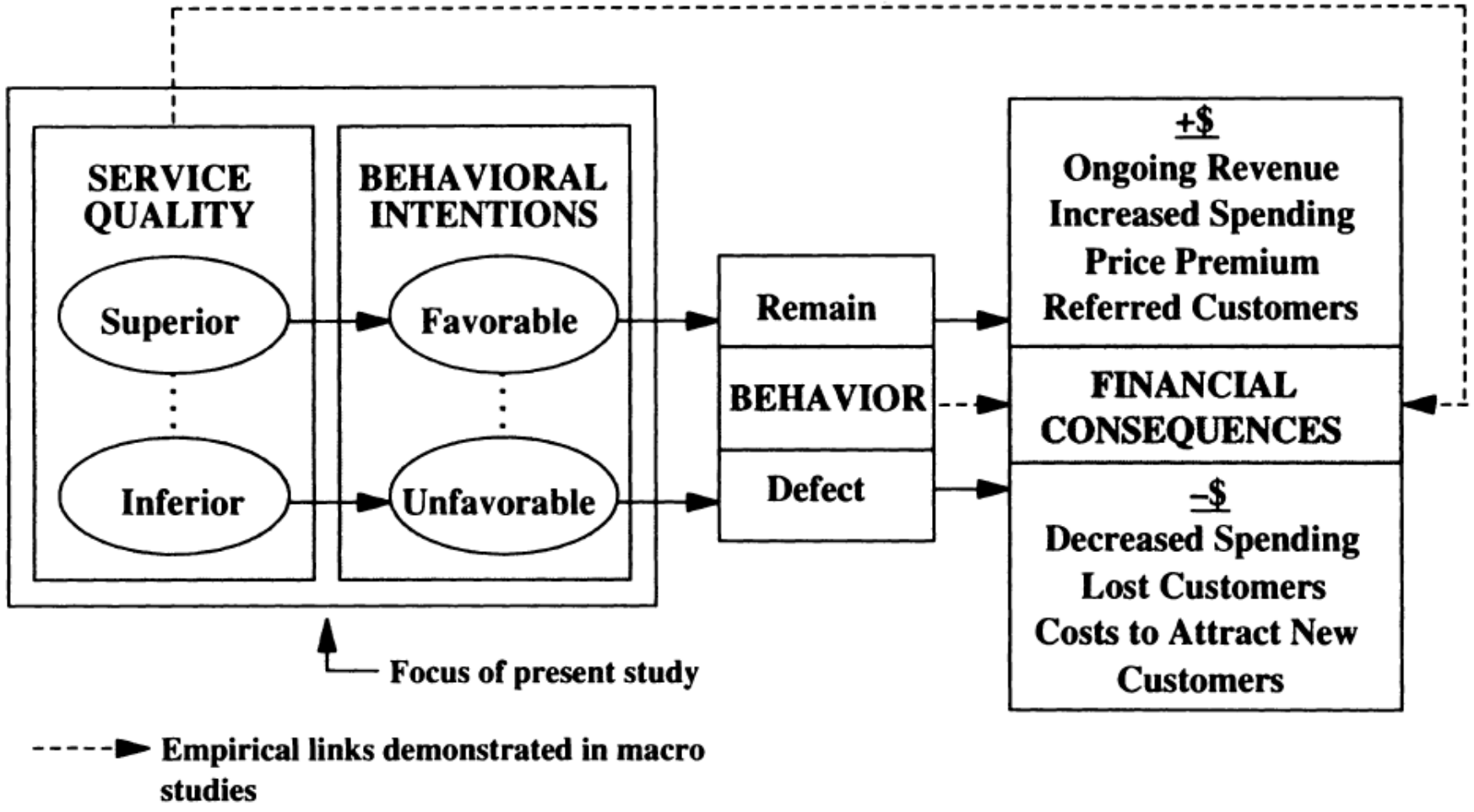


Source: Parasuraman *et al.* (1985)

Dimensions of ServQual

- Tangibles: Physical evidence of the service
- Reliability: Involves consistency of performance and dependability
- Responsiveness: Willingness or readiness of employees to provide service
- Communication: Keeping customers informed in language they can understand and listening to them
- Credibility: Trustworthiness, believability and honesty
- Security: Freedom from danger, risk or doubt
- Competence: Possession of required skills and knowledge to perform the service
- Courtesy: Politeness, respect, consideration, and friendliness of contact personnel
- Understanding/knowing the customer: Making an effort to understand customer’s needs
- Access: Approachability and ease of contact

Behavioral and Financial Linkages of ServQual



From the Author's Perspective



Understanding customer's expectations of quality

- What is the nature of customer's service expectations?
- Are there different types of expectations?
- What factors influence the formation of these expectations?
- How stable are these expectations? Do they change over time? Do they vary across service situations and across customers?
- How can companies manage expectations to enhance customer's perceptions of a service?
- What can companies' do to exceed customer's expectations?

03

The economics of quality

Crosby: "Quality is free. It's not a gift, but it is free." Is he right?

Cost of quality: pay a little early or a lot later

CONCEPT

Prevention

Stop defects being created

Non-IT: Process design, FMEA, training, poka-yoke

IT: Coding standards, threat modelling, developer training

Appraisal

Find defects before the customer does

Non-IT: Incoming inspection, audits, lab tests

IT: Code review, test automation, UAT

Internal failure

Defects caught inside the firm

Non-IT: Scrap, rework, downtime, re-inspection

IT: Bug fixes before release, failed builds

External failure

Defects that reach the customer

Non-IT: Recalls, warranty, lost customers, fines

IT: Outages, SLA penalties, breach costs, churn

Cost of good quality | **Cost of poor quality (Crosby's 'price of non-conformance')**

The '1-10-100' heuristic: ₹1 to prevent, ₹10 to catch internally, ₹100 once the customer finds it

Sources: Feigenbaum (1956) Total quality control, HBR; Crosby (1979); ASQ Cost of Quality; 1-10-100 rule attributed to Labovitz & Chang (1992). Ratios are a heuristic, not a law.

Sort the costs: prevention, appraisal or failure?

ACTIVITY · 6 MIN · TABLES

1 Training call-centre agents on a new complaint script	**2** Peer code review before every merge	**3** Re-stitching a faulty garment batch before shipment	**4** Replacing a phone under warranty
5 Running the nightly regression test suite	**6** Annual supplier quality audit	**7** Refunds and SLA penalties after a production outage	**8** Failure-mode analysis (FMEA) during product design
9 Scrapping a failed batch of tablets at a pharma plant	**10** Hot-fixing a bug found in UAT, before go-live	**11** Poka-yoke fixture that stops wrong-way assembly	**12** Customers who leave after one-star reviews

Write P / A / I / E for each card. Then: which ONE item would you cut first, and which would you double?

Answer key: where organisations actually spend

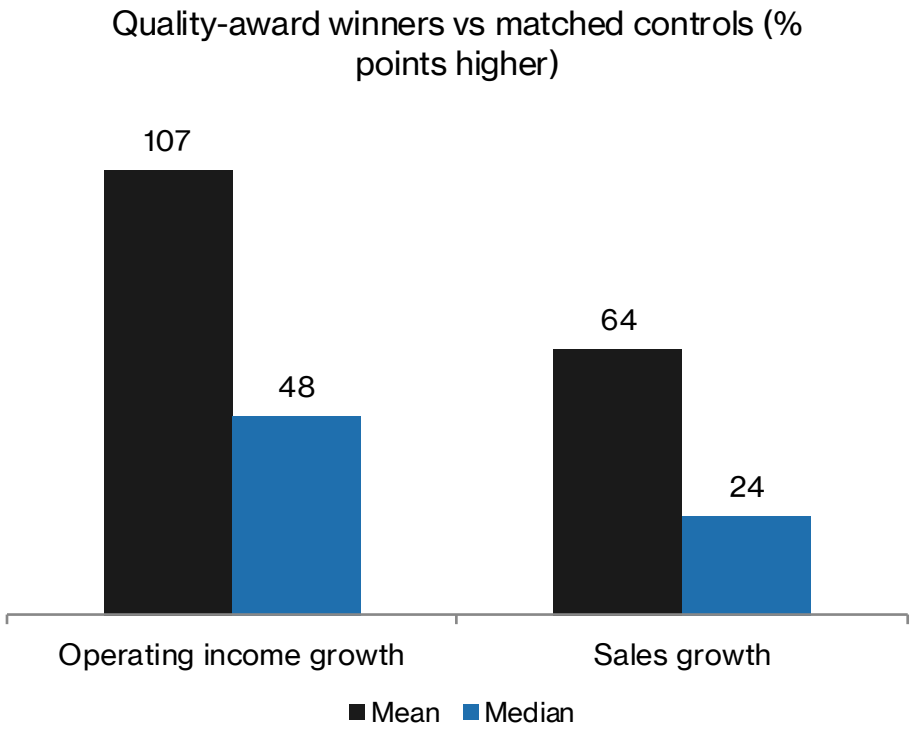
REVEAL

1 Training call-centre agents on a new complaint script **PREVENTION**	**2** Peer code review before every merge **APPRAISAL**	**3** Re-stitching a faulty garment batch before shipment **INTERNAL FAILURE**	**4** Replacing a phone under warranty **EXTERNAL FAILURE**
5 Running the nightly regression test suite **APPRAISAL**	**6** Annual supplier quality audit **APPRAISAL**	**7** Refunds and SLA penalties after a production outage **EXTERNAL FAILURE**	**8** Failure-mode analysis (FMEA) during product design **PREVENTION**
9 Scrapping a failed batch of tablets at a pharma plant **INTERNAL FAILURE**	**10** Hot-fixing a bug found in UAT, before go-live **INTERNAL FAILURE**	**11** Poka-yoke fixture that stops wrong-way assembly **PREVENTION**	**12** Customers who leave after one-star reviews **EXTERNAL FAILURE**

Typical pattern in immature organisations: heavy spending on appraisal and failure, thin on prevention. Mature ones shift the mix left.

Does quality pay? What three decades of research say

EVIDENCE



Hendricks & Singhal (1997), Management Science: ~600 award winners, 10-year window

PAYS

Levine & Toffel (2010), Mgmt Science

ISO 9001 adopters in ~1,000 California firms: far lower death rates; faster growth in sales, jobs and pay than matched non-adopters.

PAYS

Harter, Krishnan & Slaughter (2000), Mgmt Science

Higher software process maturity → higher quality; quality gains offset the extra effort of maturity.

PAYS

Swink & Jacobs (2012), J. Operations Mgmt

Six Sigma adoption linked to ROA gains, mainly via lower indirect costs.

BUT

Powell (1995), Strategic Mgmt Journal

Tools (benchmarking, training) don't create advantage; tacit culture, empowerment and executive commitment do.

BUT

Benner & Tushman (2003), AMR

Process management boosts exploitation but can crowd out exploration and radical innovation.

Full citations in References. Caution: a 2006 Fortune analysis found 91% of 58 large firms announcing Six Sigma trailed the S&P 500 afterwards; announcing a programme is not the same as building a capability.

04

The toolkits

TQM, Lean, Six Sigma, standards and DevOps: different dialects of the same grammar.


LSSa
CONTINUOUS
IMPROVEMENT
HISTORY

TQM: six principles, one hard-to-copy core

CONCEPT

Customer focus

Quality is defined by the customer, not the spec sheet

Leadership commitment

Executives own quality; it is not delegated to a QA department

Employee empowerment

People closest to the work solve problems (quality circles, andon)

Process thinking

Manage the process and the results follow; y = f(x)

Continuous improvement

Kaizen: small, frequent, everyone-every-day improvements

Fact-based decisions

Measure, then act; separate signal from noise

Powell's verdict

"...tacit, behavioral, imperfectly imitable features, such as open culture, employee empowerment, and executive commitment, can produce advantage."

The red cards are where TQM succeeds or fails. Everything else is a downloadable template.

Sources: Powell, T. C. (1995). Strategic Management Journal, 16(1), 15–37; Sousa & Voss (2002). Quality management re-visited. Journal of Operations Management, 20(1), 91–109.

Lean and TPS: eliminate waste, stop the line to fix

CONCEPT



Jidoka and andon

Any worker can stop the line when an abnormality appears. Build quality in; don't inspect it in.



Just-in-time

Make only what is needed, when needed. Low inventory exposes problems.



Kaizen

Continuous, incremental improvement by the people doing the work.



Poka-yoke

Mistake-proofing (Shingo): design so the error cannot happen.

Ohno's seven wastes, translated for IT

Waste (factory)	Equivalent in software and services
Transport	Hand-offs between teams
Inventory	Unreleased code; WIP tickets
Motion	Context switching
Waiting	Approvals, environments
Overproduction	Features nobody uses
Over-processing	Gold-plating; extra sign-offs
Defects	Bugs, rework, incidents

Sources: Ohno, T. (1988). Toyota Production System; Womack, Jones & Roos (1990); Shingo, S. (1986). Zero Quality Control; Poppendieck & Poppendieck (2003). Lean Software Development.

Six Sigma: reduce variation until defects become rare

CONCEPT

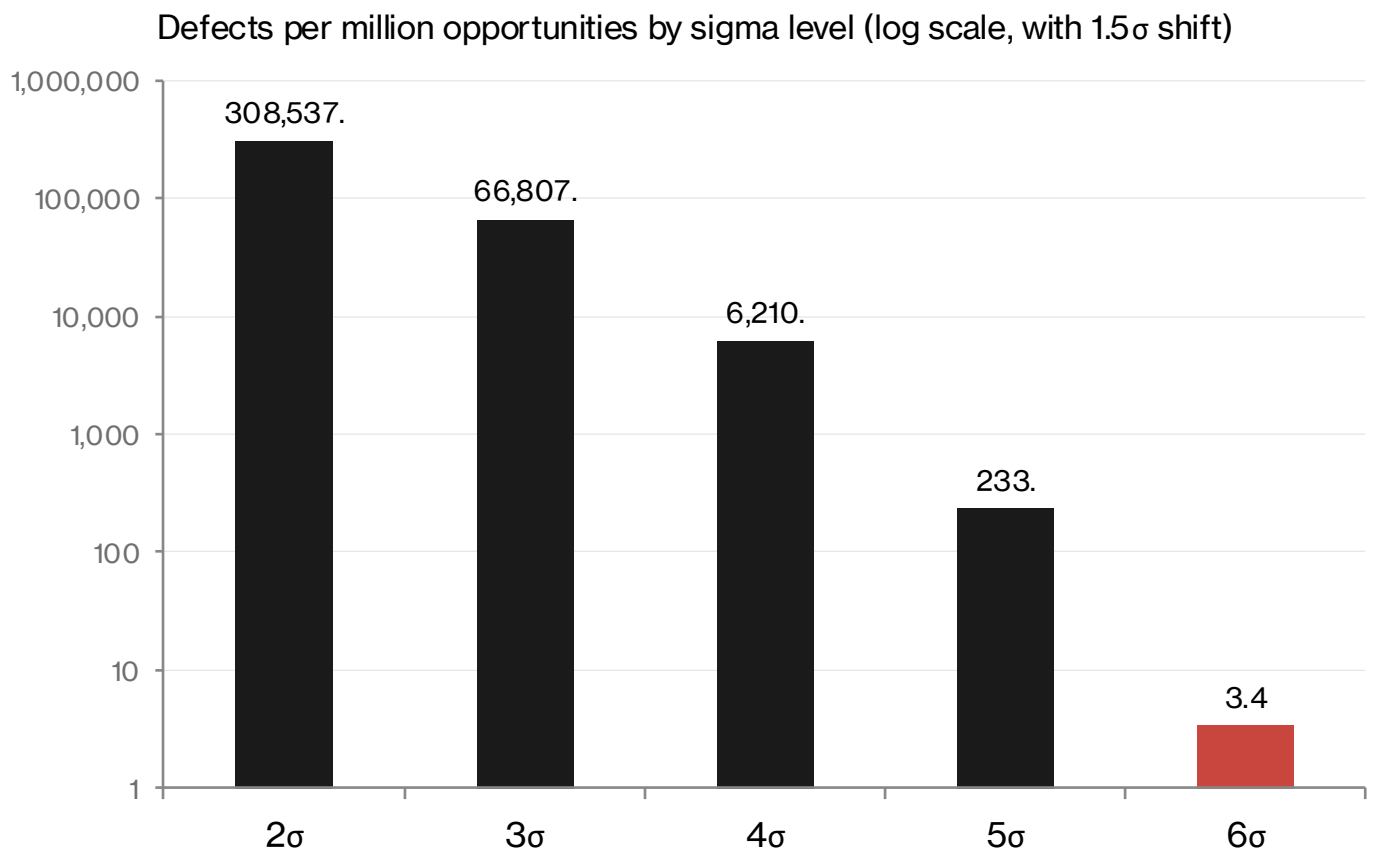
D — Define
Problem, customer CTQs, scope

M — Measure
Baseline the process and its sigma

A — Analyze
Find root causes (y = f(x))

I — Improve
Pilot and implement fixes

C — Control
Sustain with control plans and SPC



Sources: Six Sigma introduced at Motorola by Bill Smith (1986); made central to GE's strategy by Jack Welch (1995). Linderman et al. (2003), J. Operations Mgmt; Swink & Jacobs (2012).

Ishikawa's seven basic tools solve most quality problems

CONCEPT

Fishbone diagram

Brainstorm root causes by category

Check sheet

Collect defect data at source

Histogram

See the shape of variation

Pareto chart

Find the vital few causes

Control chart

Signal vs noise over time

Scatter diagram

Test a suspected x→y relationship

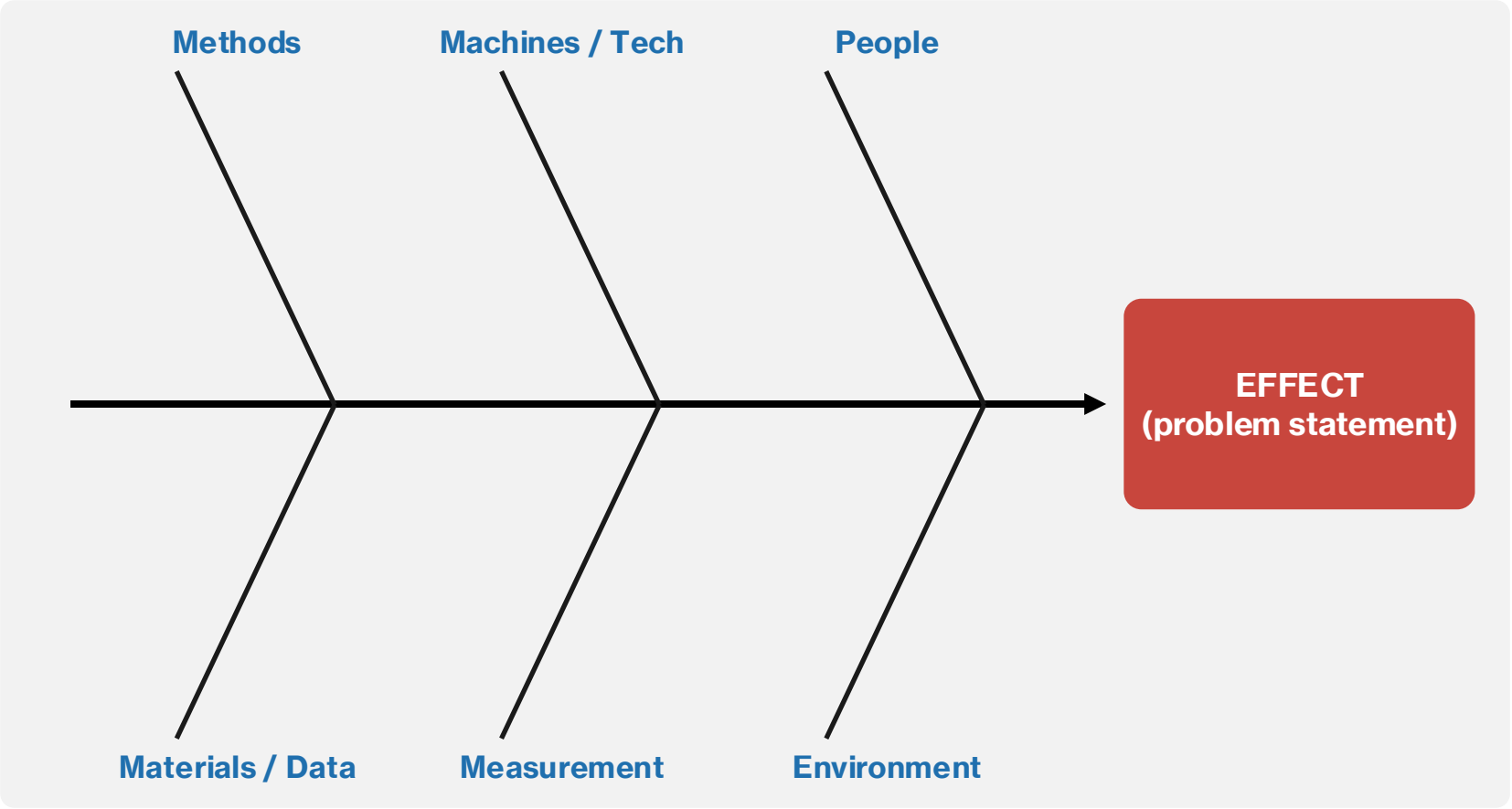
Stratification

Split data by shift, region, product, release

Source: Ishikawa, K. (1985). What Is Total Quality Control? The Japanese Way. Ishikawa held that these seven tools could resolve the large majority of workplace quality problems.

Build a fishbone: find root causes, not culprits

ACTIVITY · 12 MIN · GROUPS OF 5



Pick one effect

Non-IT: "30% of a food-delivery platform's orders in Hyderabad arrive more than 40 minutes late."

IT: "1 in 4 releases of a bank's mobile app causes a P1 incident."

≥ 3 causes per bone. Circle the 3 you would test first, and say how you'd measure them.

Pareto in 4 minutes: where should a bank's quality team start?

ACTIVITY · 4 MIN · INDIVIDUAL

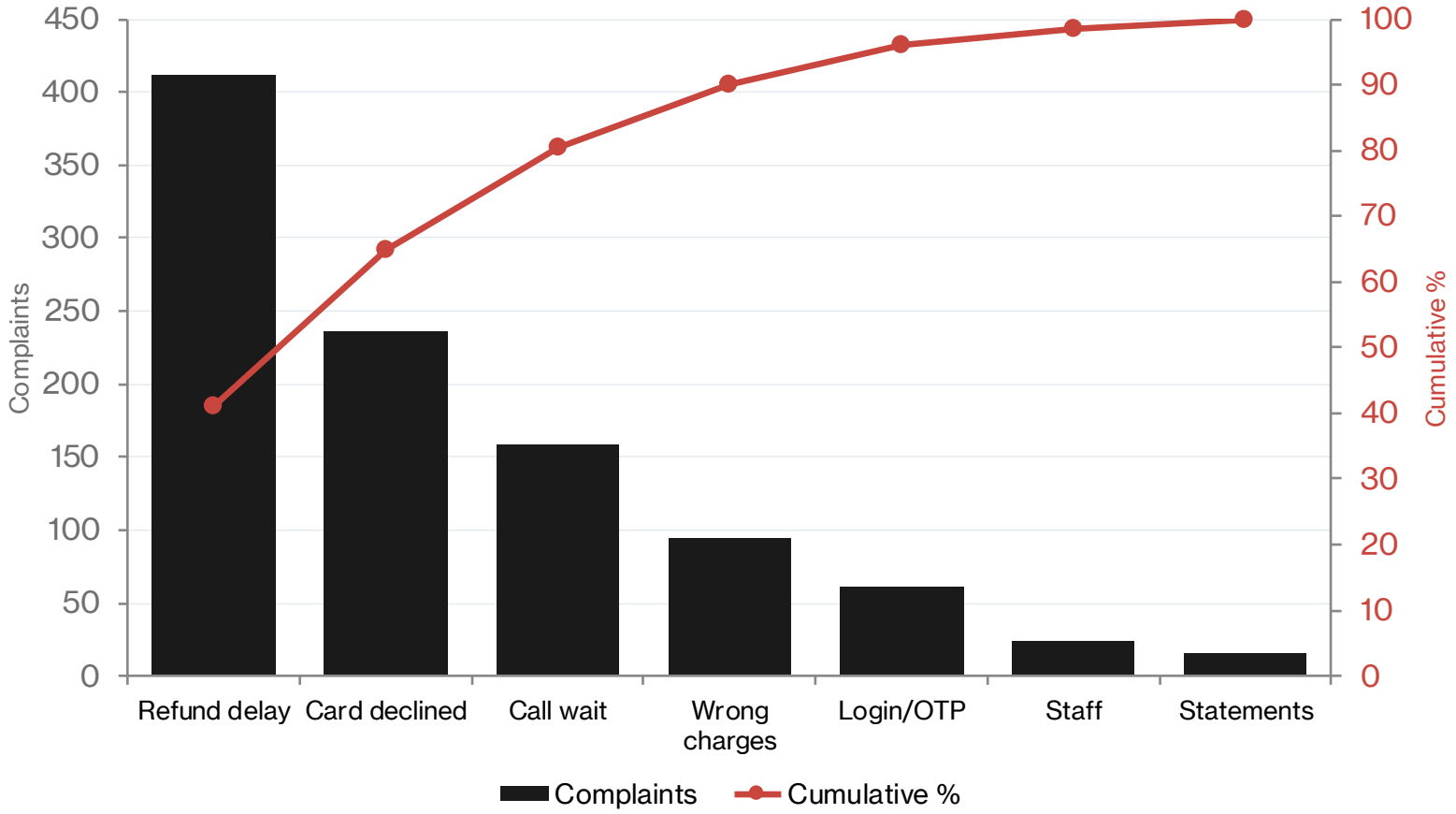
Complaint category (one month, illustrative)	Count
Delayed refund of failed UPI/IMPS	412
Card wrongly declined	236
Long wait on call centre	158
Incorrect charges	94
App login / OTP issues	61
Branch staff behaviour	24
Statement errors	15
Total	**1,000**

Your task

1. Compute the cumulative %.
2. How many categories account for ~80% of complaints?
3. Is the most frequent category also the most important? What else would you weigh (cost, regulatory risk, customer value)?

Reveal: three categories drive ~81% of complaints

REVEAL



80.6%

of complaints come from 3 of 7 categories

Juran's 'vital few vs useful many'. Fix refund automation, card-decline rules and call routing first.

Data are illustrative. Pareto principle as applied to quality: Juran, J. M. (1951). Quality Control Handbook.

Codifying quality: standards, maturity models and awards

CONCEPT

CMMI maturity levels (software and services)



ISO 9001:2015: seven quality management principles

- Customer focus
- Leadership
- Engagement of people
- Process approach
- Improvement
- Evidence-based decisions
- Relationship management

Excellence awards
Deming Prize (Japan, 1951) · Malcolm Baldrige (US, 1987) · EFQM Excellence Model (Europe) · Tata Business Excellence Model (India, Baldrige-based) · CII-EXIM Bank Award (India)

Sources: ISO 9000:2015 / ISO 9001:2015; CMMI Institute (ISACA); Deming Prize Committee (JUSE); NIST Baldrige Performance Excellence Program.

Digital quality: from QA gate to engineered reliability

CONCEPT · IT



Shift left

Move testing and security into design and coding: automated tests, code review, CI on every commit.



DORA's four keys

Deployment frequency, lead time for changes, change failure rate, time to restore. Speed and stability rise together in top performers.



SLOs and error budgets

Google SRE: a 99.9% monthly availability SLO = ~43 minutes of allowed downtime. Budget spent → freeze features, fix reliability.



AI-assisted code

DORA 2024 (39,000+ respondents): higher AI adoption associated with a 7.2% drop in delivery stability; 39% report little or no trust in AI code.

Sources: Forsgren, Humble & Kim (2018). Accelerate; Google Cloud DORA (2024). Accelerate State of DevOps Report; Beyer et al. (2016). Site Reliability Engineering, O'Reilly.

05

Cases: India and global, IT and non-IT

Nine stories, one question: was quality designed into the system or bolted on?

Case map: nine stories across two axes

CASES



India · non-IT

- Mumbai dabbawalas: precision without technology
- Aravind & Narayana: high quality at low cost
- Deming Prize leaders vs Ranbaxy



India · IT

- Indian IT services: process maturity as learning
- UPI: reliability at national scale (2025 outages)



Global · non-IT

- Toyota: TPS and the 2009–10 recall crisis
- Boeing: the 737 MAX-9 door plug (2024)



Global · IT

- CrowdStrike: 8.5 million machines down (2024)
- Knight Capital: $460 million in 45 minutes (2012)

Each case slide ends with a 'quality lens' linking it to the frameworks covered so far.

Mumbai dabbawalas: quality built from people, codes, trains

CASE · INDIA · NON-IT

~5,000
self-managed workers

~130,000
lunches delivered daily

0
smartphones or IT systems required

Organisation	Flat, autonomous units of ~25 workers; clear ownership of each route
Management	Train timetables pace the work (a natural takt time); buffer capacity and cross-training cover absences
Process	Standardised boxes and a simple colour-and-symbol code enable fast, error-proof sorting (poka-yoke)
Culture	Democratic decisions, shared identity and strong social norms make every worker an inspector

Quality lens: conformance and reliability through standardisation, mistake-proofing and culture, not technology. Ishikawa and Powell in one tiffin.

Source: Thomke, S. (2012). Mumbai's models of service excellence. Harvard Business Review, 90(11), 121–126.

Two faces of Indian manufacturing quality

CASE · INDIA · NON-IT

The Deming Prize leaders

- Sundaram Clayton (TVS group) became the first Indian Deming Prize winner in 1998.
- 2000–2012: Indian organisations won 19 Deming Prizes vs 21 for all other non-Japanese countries combined.
- 2012: all three Deming Grand Prize winners were Indian (Tata Steel, Rane Madras, Lucas-TVS).
- Clustered in auto components (TVS, Rane, Mahindra, Sundram Fasteners): suppliers to Japanese OEMs such as Maruti Suzuki.

Ranbaxy, 2013

- Pleaded guilty in the US to felony charges over adulterated drugs from its Paonta Sahib and Dewas plants.
- US$500 million in criminal fines, forfeiture and civil settlements, then the largest such case against a generic drug maker.
- Failures: incomplete stability testing, false test dates reported to the FDA, late reporting of failed batches.
- A data-integrity failure is a quality failure: appraisal that cannot be trusted is worse than none.

Sources: Wikipedia/JUSE Deming Prize winners list; W. Edwards Deming Institute (2012), '2012 Deming Prize winners'; US Department of Justice press release, 13 May 2013.

Indian IT: process maturity pays when it drives learning

CASE · INDIA · IT

Ramasubbu, Mithas, Krishnan & Kemerer (2008), MIS Quarterly: 42 projects at a CMM Level-5 offshore vendor with centres in India and the US

−1.5%

quality for every +1% in work dispersion across sites

−0.7%

productivity for every +1% in work dispersion

+0.9%

quality for every +1% in process-based learning investment

+0.6%

productivity for every +1% in learning investment

Harter, Krishnan & Slaughter (2000), Management Science
Higher CMM maturity → higher product quality. Better quality then cuts cycle time and effort enough to outweigh the overhead of maturity: quality is (nearly) free in software too.

Quality lens: CMMI was India's market signal in the 1990s–2000s; its real value is as learning infrastructure that offsets the quality penalty of distributed work.

Sources: Ramasubbu et al. (2008). MIS Quarterly, 32(2), 437–458; Harter, Krishnan & Slaughter (2000). Management Science, 46(4), 451–466.

UPI: one participant's non-conformance, a national outage

CASE · INDIA · IT

21.6 bn

UPI transactions in Dec 2025 (~698 mn/day)

~5 hrs

disruption on 12 April 2025, the longest in three years

30s → 10–15s

API response-time limits tightened from 16 June 2025

What happened

NPCI's root-cause analysis: some PSP banks flooded the switch with 'Check Transaction' status requests, far more often than the permitted once per 90 seconds, overloading shared infrastructure.

Response

Offending feature halted; rate limits enforced; faster response-time SLAs; caps on non-financial calls such as balance enquiries (50/day per app).

Quality lens

System quality = the weakest participant's conformance. Rate limits and API contracts are poka-yoke for ecosystems; ISO's 'relationship management' principle in action.

Sources: NPCI data via Elets BFSI (Jan 2026); Goodreturns, 'Understanding the recent UPI outages' (2025); The Tribune, 'UPI transactions set to become faster from June 16' (2025).

Toyota: the gold standard that grew faster than its people

CASE · GLOBAL · NON-IT

The system

- TPS gave the world jidoka, andon, just-in-time and kaizen.
- Any worker could pull the andon cord and stop the line.
- MIT's study of Toyota became 'Lean' (Womack, Jones & Roos, 1990).
- TPS was taught to suppliers through long mentoring relationships, which made it hard to scale fast.

The crisis, 2009–10

- ~8.5 million vehicles recalled for unintended-acceleration issues (floor mats, sticky pedals).
- CEO Akio Toyoda testified to the US Congress (Feb 2010).
- 2014: US$1.2 billion settlement with the US Department of Justice.

In his own words

"We pursued growth over the speed at which we were able to develop our people and our organization."
Akio Toyoda, Feb 2010

Quality lens: quality capability is built in people and scales slowly; outrun it and the system decays.

Sources: WBUR/AP (24 Feb 2010), 'Toyota CEO apologizes'; Liker & Ogden (2011). Toyota Under Fire; US DOJ (2014) deferred prosecution agreement.

Boeing 737 MAX-9: when four missing bolts reveal a culture

CASE · GLOBAL · NON-IT

5 Jan 2024 — Alaska Airlines Flight 1282: a door plug blows out shortly after take-off from Portland. Eight minor injuries; no fatalities.

Jan 2024 — FAA grounds 171 MAX-9s and caps 737 MAX production at 38 per month.

Jun 2025 — NTSB: the four bolts securing the plug were never installed at Boeing's factory. It cites inadequate training, guidance and oversight, and FAA audit shortcomings.

Oct 2025 — FAA lifts the cap to 42/month after ~21 months: a regulator using throughput as the quality lever.

Quality lens

- Appraisal failed: inspection did not catch missing parts.
- Prevention failed: no reliable process to record when the plug was opened and refitted.
- Context: two MAX-8 crashes (2018, 2019) killed 346 people; investigators repeatedly pointed to culture and schedule pressure.

Debate: should production rate be a quality-control variable?

Sources: NTSB final report on Alaska Airlines Flight 1282 (2025), via AVweb and Seattle Times; CNBC (17 Oct 2025), 'FAA raises Boeing 737 Max production cap to 42 a month'.

Software at scale: one bad change, global blast radius

CASE · GLOBAL · IT

CrowdStrike, 19 July 2024

8.5 mn
Windows devices crashed

78 min
until the update was reverted

~$5.4 bn
est. losses for Fortune 500 firms

A configuration update (Channel File 291) to the Falcon sensor caused an out-of-bounds memory read and boot loops. Airlines, banks and hospitals worldwide were hit; Delta alone cited ~US$500 million. Fixes: staged (canary) rollouts and stronger content validation.

Knight Capital, 1 August 2012

45 min
of runaway trading

1 of 8
servers missed the new code

$460 mn+
loss; firm sold within months

A repurposed feature flag woke up dormant 'Power Peg' code on the one server not updated. The SEC found no written deployment procedure, no second-person review and no regression testing of dead code.

Quality lens: both were deployment-process failures, not 'one bad developer'. Small batches, canaries, two-person review and automated checks are the poka-yoke of software.

Sources: Wikipedia, '2024 CrowdStrike-related IT outages' (Parametrix estimate); SEC (2013) In the Matter of Knight Capital Americas LLC, via J. R. Varma's analysis.

06

Synthesis

What should a manager take away on Monday morning?

Seven ideas to carry out of the room

SYNTHESIS

1. **Choose your definition**

   Quality is multi-dimensional; strategy is choosing which dimensions to win on (Garvin).

2. **Fix the system, not the person**

   Most variation is common cause; blame is usually tampering (Deming).

3. **Prevention is cheapest**

   Every rupee moved upstream saves many downstream (Feigenbaum, Crosby).

4. **Scale changes everything**

   99.9% is a crisis at 700 million transactions a day.

5. **Culture beats toolkits**

   Tools are imitable; empowerment and leadership commitment are not (Powell).

6. **Quality and cost can move together**

   Aravind, Narayana and Toyota show the trade-off is often a design choice.

7. **Mind the exploration trap**

   Optimise processes without killing innovation (Benner & Tushman).

Exit ticket: 3-2-1 before you leave

ACTIVITY · 5 MIN · INDIVIDUAL

3

concepts from today you could explain to a fellow student on campus tomorrow

2

quality costs (P-A-I-E) you have seen in your own organisation or internship

1

question you still have, or one system you would redesign

References

SOURCES

Benner, M. J., & Tushman, M. L. (2003). Exploitation, exploration, and process management. Academy of Management Review, 28(2), 238–256.

Crosby, P. B. (1979). Quality Is Free. McGraw-Hill.

Deming, W. E. (1986). Out of the Crisis. MIT CAES.

Feigenbaum, A. V. (1956). Total quality control. Harvard Business Review, 34(6), 93–101.

Forsgren, N., Humble, J., & Kim, G. (2018). Accelerate. IT Revolution.

Garvin, D. A. (1984). What does "product quality" really mean? Sloan Management Review, 26(1), 25–43.

Garvin, D. A. (1987). Competing on the eight dimensions of quality. Harvard Business Review, 65(6), 101–109.

Google Cloud DORA (2024). Accelerate State of DevOps Report 2024.

Govindarajan, V., & Ramamurti, R. (2013). Delivering world-class health care, affordably. Harvard Business Review, 91(11).

Haripriya, A., et al. (2012). Complication rates of phacoemulsification and MSICS at Aravind Eye Hospital. J. Cataract & Refractive Surgery, 38(8).

Harter, D. E., Krishnan, M. S., & Slaughter, S. A. (2000). Effects of process maturity on quality, cycle time, and effort in software product development. Management Science, 46(4), 451–466.

Hendricks, K. B., & Singhal, V. R. (1997). Does implementing an effective TQM program actually improve operating performance? Management Science, 43(9), 1258–1274.

Ishikawa, K. (1985). What Is Total Quality Control? The Japanese Way. Prentice-Hall.

Juran, J. M. (1986). The quality trilogy. Quality Progress, 19(8), 19–24.

Levine, D. I., & Toffel, M. W. (2010). Quality management and job quality: How the ISO 9001 standard affects employees and employers. Management Science, 56(6), 978–996.

Liker, J. K., & Ogden, T. N. (2011). Toyota Under Fire. McGraw-Hill.

NTSB (2025). Aircraft Accident Report: Alaska Airlines Flight 1282.

Ohno, T. (1988). Toyota Production System: Beyond Large-Scale Production. Productivity Press.

Powell, T. C. (1995). Total quality management as competitive advantage. Strategic Management Journal, 16(1), 15–37.

Ramasubbu, N., Mithas, S., Krishnan, M. S., & Kemerer, C. F. (2008). Work dispersion, process-based learning, and offshore software development performance. MIS Quarterly, 32(2), 437–458.

SEC (2013). In the Matter of Knight Capital Americas LLC. Release No. 70694.

Sousa, R., & Voss, C. A. (2002). Quality management re-visited. Journal of Operations Management, 20(1), 91–109.

Swink, M., & Jacobs, B. W. (2012). Six Sigma adoption: Operating performance impacts and contextual drivers of success. Journal of Operations Management, 30(6), 437–453.

Thomke, S. (2012). Mumbai's models of service excellence. Harvard Business Review, 90(11), 121–126.

US Department of Justice (2013, May 13). Generic drug manufacturer Ranbaxy pleads guilty and agrees to pay $500 million.